

KEPLER: Knowledge Extraction Pipeline for Logical Evidence and Reasoning

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1 Motivation

Misinformation spreads rapidly across text and images, eroding trust in public discourse. Manual fact-checking cannot scale to the volume and velocity of claims; automated approaches typically focus on single modalities, rely on fixed corpora, or lack transparent justifications. **KEPLER** addresses these gaps through a real-time, multimodal pipeline that couples robust evidence retrieval with logical, multi-hop reasoning and confidence-rated verdicts for end users (journalists, researchers, platforms).

2 Related Work

Early pipelines (e.g., FEVER (Thorne et al., 2018)) emphasize text-only verification; multi-hop datasets (e.g., HOVER (Jiang et al., 2020), EX-FEVER) advance reasoning with explainable 2-hop and 3-hop chains (Ma et al., 2024), yet most systems remain limited to fixed text corpora. Recent multimodal LLMs (e.g., GPT-4o, LLaVA) enable image+text comprehension, and DEFAME (Braun et al., 2025) demonstrates dynamic evidence-based processing for multimodal claims. However, these approaches lack systematic integration into end-to-end fact-checking with real-time retrieval, calibrated uncertainty quantification, and structured justifications. **KEPLER** contributes a cross-modal, model-agnostic backbone with iterative retrieval, explicit confidence estimation, and transparent reasoning chains.

3 Proposed Methods and Experiments

Pipeline. (1) *Input Handler & Claim Detection Agent* Accepts both text and image inputs. Utilizes user-selected multimodal LLMs, allowing the use of one or multiple models (e.g., GPT-4o, Gemini, Claude) to parse and identify distinct, verifiable factual claims for further verification. (2) *Retriever Agent* Designs tailored ev-

idence search strategies based on claim modality. Calls APIs (Google Search, GeoClip, Vision) to collect fresh, relevant content from unbiased sources. (3) *Reranker Agent* Filters and prioritizes retrieved evidence. Orders are sourced by relevance, credibility, and recency to ensure only the strongest and most pertinent facts move forward. (4) *Aggregator Agent* Synthesizes multimodal evidence by integrating text, images, and metadata. It applies chain-of-thought reasoning and highlights agreements, conflicts, and missing information. (5) *Verifier Agent* Evaluates the aggregated evidence for each claim. Assigns verdicts (Supported, Refuted, Misleading, Not Enough Information), with iterative evidence expansion for uncertain cases. (6) *Weighted Multi-Model Output* When multiple LLMs are selected, their individual verdicts, explanations, and confidence scores are aggregated using a dynamic weighting system. This ensures robust, balanced fact-checking decisions, capitalizing on the strengths and diversity of each model. (7) *Confidence Scorer* Rates verdict certainty by considering source reliability, cross-model agreement, and evidence recency. Presents confidence scores and structured, source-linked justifications.

Experiments. Benchmark performance on standard datasets (FEVER (Thorne et al., 2018), HOVER (Jiang et al., 2020), AVeriTeC (Schlichtkrull et al., 2023), MOCHEG (Yao et al., 2023), VERITE (Papadopoulos et al., 2024), PUBHEALTH (Kotonya and Toni, 2020)), measuring claim-level accuracy, precision, recall, F1, and explanation quality (BLEU, METEOR, ROUGE).

Metrics. Assess calibration using Brier Score and Expected Calibration Error.

4 Plan and Milestones

- **Weeks 1–2:** Literature review; finalize datasets/splits; claim extractor.
- **Weeks 3–6:** Evidence planner; retrieval API integration (text/image/geo).
- **Weeks 7–10:** Summarization; contrastive reasoning logic; uncertainty.
- **Weeks 11–13:** Verdicts; NEI re-loop; justification & UI.
- **Week 14:** Benchmarking vs. baselines; human study; write-up.

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